

SCIENCE — Q.P. Code 31/1/3 (Set ~3)

Section A — Questions 9 to 17 (Multiple Choice, 1 mark each)

9. If pea plants with round and green seeds (RRyy) are crossed with pea plants having wrinkled and yellow seeds (rrYY), the seeds developed by the plants of F_1 generation will be :

- (A) 50% round and green
- (B) 75% wrinkled and green
- (C) 100% round and yellow
- (D) 75% wrinkled and yellow

10. The correct/true statement(s) for a bisexual flower is/are :

- (i) They possess both stamen and pistil.
- (ii) They possess either stamen or pistil.
- (iii) They exhibit either self-pollination or cross-pollination.
- (iv) They cannot produce fruits on their own.

- (A) (i) only
- (B) (iv) only
- (C) (i) and (iii)
- (D) (i) and (iv)

11. The plant hormone whose concentration stimulates the cells to grow longer on the side of the shoot which is away from light is :

- (A) Cytokinins
- (B) Gibberellins
- (C) Adrenaline
- (D) Auxins

12. Secretion of less saliva in mouth will effect the conversion of :

- (A) proteins into amino acids
- (B) fats into fatty acids and glycerol
- (C) starch into simple sugars
- (D) sugars into alcohol

13. The percentage of solar energy which is not converted into food energy by the leaves of green plants in a terrestrial ecosystem is about :

- (A) 1%
- (B) 10%
- (C) 90%
- (D) 99%

14. Which of the following groups do not constitute a food chain ?

- (i) Wolf, rabbit, grass, lion
- (ii) Plankton, man, grasshopper, fish
- (iii) Hawk, grass, snake, grasshopper, frog
- (iv) Grass, snake, wolf, tiger

- (A) (i) and (iv)
- (B) (i) and (iii)
- (C) (ii) and (iii)
- (D) (ii) and (iv)

15. The phenomenon responsible for making the smoke particles visible when a beam of sunlight enters a smoke filled room through a narrow hole is :

- (A) scattering of light
- (B) dispersion of light
- (C) reflection of light
- (D) internal reflection of light

16. Mirror 'X' is used to concentrate sunlight in solar furnace and Mirror 'Y' is fitted on the side of the vehicle to see the traffic behind the driver. Which of the following statements are true for the two mirrors ?

- (i) The image formed by mirror 'X' is real, diminished and at its focus.
- (ii) The image formed by mirror 'Y' is virtual, diminished and erect.
- (iii) The image formed by mirror 'X' is virtual, diminished and erect.
- (iv) The image formed by mirror 'Y' is real, diminished and at its focus.

- (A) (i) and (ii)
- (B) (ii) and (iii)
- (C) (iii) and (iv)
- (D) (i) and (iv)

For Questions number 17 to 20, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
- (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
- (C) Assertion (A) is true, but Reason (R) is false.
- (D) Assertion (A) is false, but Reason (R) is true.

17. **Assertion (A) :** The amount of ozone in the atmosphere began to drop sharply in the 1980s.

Reason (R) : The oxygen atoms combine with molecular oxygen to form ozone.